Source 1:
Application of dimensionality reduction and clustering algorithms for the classification of kinematic morphologies of galaxies

- Uses UMAP to transform high-dimensional data (~900 or ~2700 features) into a low-dimensional form (2 dimensions) before feeding these reduced dimensions into HDBSCAN
    - I'm not entirely sure whether 900 features or 2700 features were used. I couldn't follow the explanation provided by the paper. This part can be found on the left side of page 5 if it's relevant
    - HDBSCAN is a type of density-based clustering algorithm
        - The algorithm tries to find dense clusters of points and then groups these clusters together
        - HDBSCAN is a variation of DBSCAN that's been modified to be less argument-dependent
            - DBSCAN does poorly if a bad ε value is chosen
            - ε represents the maximum radius around a data point for another data point to be considered its "neighbor"

Source 2:
Wide Area VISTA Extra-galactic Survey (WAVES): unsupervised star-galaxy separation on the WAVES-Wide photometric input catalogue using UMAP and hdbscan

- Same idea as previous, uses UMAP and then HDBSCAN
    - Reduces 48 features (or 34 features without the u-band)
    - Doesn't explicitly say how many dimensions it got reduced to after UMAP, but I would assume it reduced to 2 or 3

Source 3:
A review of unsupervised learning in astronomy

- This paper doesn't focus on any particular research and instead covers the different ways that unsupervised machine learning can be applied to astronomy (as the title suggests)
- Section 4.2.4 talks about t-SNE
    - Sounds like UMAP is better for most situations
    - Aadi tested this approach a bit, but I never looked too deep into it
        - Could be worth asking him what his thoughts are
- Section 4.2.5 talks about UMAP
    - Suggests that UMAP is better for data exploration/visualization than classification
        - Lots of reasons provided, refer to that section for more information
- Section 5.6 talks about density-based clustering algorithms
    - Also mentions applying a dimensionality reduction before using these classifiers

Finding Feature Importance with UMAP

It looks like UMAP was created by Leland McInnes. This is his GitHub. Someone in the issues section of his GitHub recommended an approach to find feature importance that seemed fairly similar to the idea that Dr. Sandquist mentioned in our meeting on 10/16/2025.

Link

The approach that they recommended followed this general pipeline

1. Fit UMAP to the original data
2. Choose a feature and randomly shuffle the values in the column (basically makes that column useless)
3. Fit UMAP to this new dataset
4. Repeat this for every feature
5. Compare how different each new UMAP variation is from the orignal